

HOMOLOGICAL METHODS IN ALGEBRAIC GEOMETRY

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1. INTRODUCTION

Algebraic geometers study a shape through invariants attached to it. Schemes, and more generally algebraic spaces, generalize shapes cut out by polynomial equations. In particular, algebraic spaces naturally arise as quotients by étale equivalence relations or group actions.

One of the richest invariants attached to an algebraic space is its derived category of quasi-coherent sheaves. An equivalence between derived categories can relate spaces with different equations while preserving subtle information. Many of these equivalences are realized by integral transforms, also called Fourier–Mukai transforms [35]. Moreover, many singularities arising in birational geometry and the minimal model program are naturally defined in derived category.

My previous research has studied how geometric and singularity-theoretic information is encoded in derived categories of schemes, algebraic spaces, and algebraic stacks. I developed and applied techniques involving t -structures, approximation, and generation for triangulated categories to the study structure of these derived categories. This work has led to classification results, bounds on invariants like Rouquier dimension, and categorical characterizations of singularities.

Currently, I have two ongoing research programs that lie at the intersection of geometry, homological algebra, K -theory, and singularity theory: *Fourier–Mukai partnership and higher singularities*. These are discussed in the following sections, and together advance our understanding for derived categories of singular spaces.

2. FOURIER–MUKAI PARTNERSHIP

Let $f_i: Y_i \rightarrow S$ be proper flat morphisms of Noetherian algebraic spaces. Denote by $p_i: Y_1 \times_S Y_2 \rightarrow Y_i$ be the projections. An object $E \in D_{\text{qc}}(Y_1 \times_S Y_2)$ determines an integral transform $\Phi_E: D_{\text{qc}}(Y_1) \rightarrow D_{\text{qc}}(Y_2)$ defined by $A \mapsto \mathbf{R}(p_2)_*(\mathbf{L}(p_1)^* A \otimes^{\mathbf{L}} E)$. If Φ_E restricts to an equivalence on D_{coh}^b , then we say Y_1 and Y_2 are Fourier–Mukai S -partners.

For $p \in |S|$, choose a representative $t: \text{Spec}(k) \rightarrow S$. Let E_p be the derived pullback of E to the corresponding fiber product and let Φ_{E_p} be the induced transform on the fibers. Define

$$\text{fm}(E) := \{p \in |S| : \Phi_{E_p} \text{ is fully faithful on } D_{\text{coh}}^b\}$$

and

$$\text{FM}(E) := \{p \in |S| : \Phi_{E_p} \text{ is an equivalence on } D_{\text{coh}}^b\}.$$

It is a natural question as to whether these sets are well-defined, independently of the chosen representative of p , and what their topology is like. This part of my work is centered on the following questions, pursued jointly with my PhD student Guisado Villalgorido:

Objective A. (Locus). *When do integral transforms with possibly nonperfect kernels preserve bounded coherent and perfect complexes, and when are the loci $\text{fm}(K)$ and $\text{FM}(K)$ open?*

Objective B. (Singularity categories). *When does an integral transform induce a fully faithful functor or an equivalence on singularity categories, and to what extent can this be detected fiberwise?*

Objective C. (Invariances). *What geometric properties of relative Fourier–Mukai partners are derived invariant? In particular, are Gorensteinness or smoothness preserved?*

2.1. State of the art. Most existing methods for relative integral transforms require the kernel to be perfect. See e.g. [18, Theorem 5.8] and [7, Proposition 4.9]. Perfectness is also built into standard constructions representing the unit and counit of an adjunction by morphisms of kernels [2]. These methods therefore do not directly cover singular families, where natural kernels may be pseudocoherent but nonperfect.

This is not a technical obstruction. To see:

Example 1. Let $k := \mathbb{F}_3(t)$ where t is transcendental. Consider the projective plane curve $f: C \rightarrow \operatorname{Spec}(k)$ given by the equation $y^2z + x^3 - tz^3 = 0$. This is a regular curve. After base change of C along the field extension $\ell := \mathbb{F}_3(t^{\frac{1}{3}})$, $C \times_k \operatorname{Spec}(\ell)$ is a singular projective curve [25, Remark 16]. Denote by $t': C \times_k C \times_k \operatorname{Spec}(\ell) \rightarrow C \times_k C$ the natural base change morphism. Choose any $K \in D_{\operatorname{coh}}^b(C \times_k C)$ for which its associated integral transform $\Phi_K: D_{\operatorname{qc}}(C) \rightarrow D_{\operatorname{qc}}(C)$ is an equivalence. For example, we can take $K := (\Delta_f)_* \mathcal{O}_C$, which is nonperfect. More generally, no kernel K inducing such an autoequivalence can be perfect. Indeed, if K were perfect, then $\mathbf{L}(t')^* K$ would be perfect. By [17, Corollary 1.3], the associated integral transform remains an equivalence after base change. However, [15, Proposition 4.6] implies that $\mathbf{L}(t')^* K$ cannot be perfect because $C \times_k \operatorname{Spec}(\ell)$ is singular, which is a contradiction.

Thus, any theory that is stable under base change and applies to singular fibers must allow nonperfect kernels. Recent work has established important ascent and descent results for schemes [17]. However, a general theory for algebraic spaces, open Fourier–Mukai loci (not yet known for schemes), and further applications remain open.

This research program builds naturally on my recent work on integral transforms, relative Fourier–Mukai theory, and categorical methods [15, 17, 32, 5].

2.2. Openness of the Fourier–Mukai locus. Various notions of relative perfectness have been studied for schemes [21, 1, 20, 6, 41]. These notions characterize when an integral transform preserves bounded pseudocoherent or perfect complexes. In our work, we clarify that the various notions of relatively perfect complexes for morphisms of schemes appearing in the literature are equivalent. This comparison appears to be missing from the literature. See e.g. the discussion in [1, §1.4] and the remarks following [19, Definition 1.3].

We introduce analogous notions for algebraic spaces. Let us define them. Let $f: Y \rightarrow X$ be a morphism of Noetherian algebraic spaces. We say that $E \in D_{\operatorname{qc}}(Y)$ is relatively perfect over X , or f -perfect, if $E \otimes^{\mathbf{L}} \mathbf{L}f^* A \in D_{\operatorname{qc}}^b(Y)$ for all $A \in D_{\operatorname{qc}}^b(X)$; E is relatively quasi-perfect over X , or f -quasi-perfect, if $\mathbf{R}f_*(E \otimes^{\mathbf{L}} P) \in \operatorname{Perf}(X)$ for all $P \in \operatorname{Perf}(Y)$.

The first notion is motivated by [1], whereas the second is inspired by [6, §3]. In general, the two notions do not coincide. For a morphism to a field, the structure sheaf is relatively perfect, but it need not be relatively quasi-perfect when the morphism is nonproper. However, our preliminary work shows that they do coincide under suitable hypotheses for proper morphisms of algebraic spaces. We prove:

Objective A. 1. Let $f_1: Y_1 \rightarrow S$ and $f_2: Y_2 \rightarrow S$ be proper morphisms of Noetherian algebraic spaces. Denote by $p_i: Y_1 \times_S Y_2 \rightarrow Y_i$ the natural projections. For any pseudocoherent $E \in D_{\operatorname{qc}}(Y_1 \times_S Y_2)$,

- (1) E is p_1 -perfect $\iff \Phi_E(D_{\operatorname{coh}}^b(Y_1)) \subseteq D_{\operatorname{coh}}^b(Y_2)$
- (2) E is p_2 -perfect $\iff \Phi_E(\operatorname{Perf}(Y_1)) \subseteq \operatorname{Perf}(Y_2)$.

Objective A. 1 is the first structural result: it characterizes when an integral transform preserves bounded pseudocoherent complexes or perfect complexes. Since $D_{\operatorname{qc}}(Y_i)$ admits a single compact generator, these preservation properties can be tested on one perfect complex. This reduction is used in **Objective A. 3** where the failure of fully faithfulness and equivalence is expressed through supports of unit and counit defects evaluated on compact generators.

A key technical ingredient is a characterization of perfect complexes on algebraic spaces, generalizing [1, Theorem 2.3]. The implication from p_1 -perfectness to preservation of bounded pseudocoherent complexes relies on detecting boundedness of cohomology via the vanishing of Ext-groups against a compact generator. The main difficulty lies in proving that relative quasi-perfectness coincides with relative perfectness.

Our strategy combines compact generation of triangulated categories with generation by thick subcategories in the sense of [8]. A key observation is that relatively quasi-perfect complexes admit natural involutions that induce useful adjunctions. In fact, these adjunctions drive the comparison between the two notions.

A guiding principle is that global categorical behavior (e.g. equivalences, fully faithful functors, and adjunctions) is controlled by fiberwise data. This philosophy already appears in [20] where it is shown that, for locally projective flat morphisms of finitely presented schemes over an algebraically closed field, fully faithfulness and equivalences of integral transforms may be detected on special fibers.

Recently, [17] extended this principle to proper flat morphisms of Noetherian schemes. Our work extends these fiberwise criteria to suitable algebraic spaces. The extension requires substantial new technical developments described below.

Our preliminary work establishes the following fiberwise criterion.

Objective A. 2. Let $f_1: Y_1 \rightarrow S$ and $f_2: Y_2 \rightarrow S$ be proper flat morphisms of Noetherian algebraic spaces. Suppose $K \in D_{\operatorname{qc}}(Y_1 \times_S Y_2)$ is pseudocoherent and relatively perfect over each Y_i . Then the following are equivalent:

(1) Φ_K is fully faithful on D_{coh}^b

(2) for any closed point $p \in |S|$, the base-changed integral transform Φ_{K_p} is fully faithful on D_{coh}^b .

Furthermore, if $\Phi_K \dashv \Phi_{K'}$ where K' is relatively perfect over Y_1 , then the analogous fiberwise criterion detects when the adjoint pair is an equivalence.

This is a space-theoretic analog of [17, Theorem 1.2], and its proof requires several new ideas. Unlike schemes, the underlying topological space of an algebraic space consists of equivalence classes of morphisms from spectra of fields. Hence, we establish compatibility with faithfully flat and affine base change, as well as independence of the chosen representative of a point.

The first ingredient is a study of relative perfectness under base change. In particular, we prove relative perfectness can be detected after faithfully flat presentations and on fibers over closed points. Crucially, such results provide the descent and ascent results needed later. The second ingredient is the construction of adjoints for integral transforms, and the proof of their compatibility with base change. Here we rely on Neeman's extension of Grothendieck duality to algebraic spaces [38].

The key new tool is a theory of derived reflexivity for algebraic spaces, which extends [3]. In particular, we show that derived reflexivity is étale local. We prove that relatively perfect complexes are naturally reflexive with respect to 'relative dualizing complexes'. These results allow us to construct explicit adjoints for integral transforms and establish their compatibility with base change.

With these ingredients, the arguments proving the fiberwise criterion largely parallel those for schemes in [17, §4]. The essential inputs are: existence and control of adjoints; compatibility of integral transforms with flat base change; detection of relative perfectness and boundedness by faithfully flat descent. Although our proof also uses mates, its strategy differs greatly from that of [7]. Their arguments rely crucially on kernels being perfect, whereas our kernels need only be pseudocoherent. In particular, the techniques of [7, Proposition 4.9] are unavailable, and the machinery developed above is essential.

The results above recover space-theoretic analogs of recent ascent and descent theorems for schemes [15, 17]. They also extend results on adjoints of integral transforms due to [6, 41]. Notably, our work yields a new proof of [17, Proposition 4.9] (and hence, [20, Proposition 2.15]), making our approach largely self-contained.

Combining [Objective A. 1](#) and [Objective A. 2](#), we can prove:

Objective A. 3. Consider proper flat surjective morphisms $f_1: Y_1 \rightarrow S$ and $f_2: Y_2 \rightarrow S$ of Noetherian algebraic spaces. For any $K \in D_{\text{coh}}^b(Y_1 \times_S Y_2)$ which is relatively perfect over each Y_i , $\text{fm}(K)$ and $\text{FM}(K)$ are open subsets of $|S|$.

In other words, [Objective A. 3](#) establishes openness of the loci of points where base-changed integral transforms are fully faithful or an equivalence. Even for schemes, this is quite new, particularly because it applies in arbitrary characteristic. This motivates the definitions of $\text{fm}(-)$ and $\text{FM}(-)$.

Our work identifies the complement of $\text{fm}(K)$ and $\text{FM}(K)$ with the image of the support of the cones of the unit or counit of an adjunction applied to compact generators. Specifically, fully faithfulness is controlled by the support of the cone of the unit on a compact generator, whereas equivalence is controlled by the union of the supports of the cones of the unit and counit for compact generators. Furthermore, we verified independence of the representative for p , which asserts that $\text{fm}(-)$ and $\text{FM}(-)$ are well-defined.

Neither this description of the loci nor the resulting openness theorem appears to be known even for schemes. This generalizes the algebraic space version of [18, Corollary 5.9] in two ways: first, we do not require perfect kernels; second, the arguments differ because those of [2] are unavailable. In fact, the techniques here are independent.

Now we apply the preceding theory to construct autoequivalences of elliptic fibrations of algebraic spaces. Recall that an *elliptic X -fibration* is a proper Gorenstein morphism $Y \rightarrow X$ between Noetherian algebraic spaces whose geometric fibers are integral curves of arithmetic genus one with trivial dualizing sheaf. Basic examples include fiber products of (perhaps singular) elliptic curves.

Preliminary work shows:

Objective A. 4. Let X be a Noetherian algebraic space. Consider an elliptic X -fibration $f: Y \rightarrow X$. Then $\Phi_{\mathcal{I}_{\Delta_f}}$ induces an autoequivalence of $D_{\text{coh}}^b(Y)$ where $\mathcal{I}_{\Delta_f}$ is the ideal sheaf of the diagonal $\Delta_f: Y \rightarrow Y \times_X Y$.

To do so, a key ingredient is studying the behavior of diagonal ideal sheaves under affine base change. This allows us to generalize [20, Proposition 2.16] to algebraic spaces. Notably, the result removes the separatedness assumptions on the source and target and applies in mixed characteristic. As an example, let S be a Dedekind scheme and $f: Y \rightarrow S$ an elliptic fibration extending an elliptic curve over the generic point. We show the base-changed integral transform with kernel $\mathcal{I}_{\Delta_f}$ along the generic fiber is an autoequivalence. Openness then

implies that the base-changed integral transform with kernel $\mathcal{J}_{\Delta_f}$ is an autoequivalence on all but finitely many closed fibers, equivalently after all but finitely many reductions modulo primes.

Objective A. 4 combines all the developed theory into a single application. In particular, the fiberwise criterion **Objective A. 2** allows equivalences to be tested after passing to algebraic closures of residual spaces. The main technical hurdle is that desired equivalences occur on D_{coh}^b . For schemes, results of weak approximability and uniqueness-of-enhancements allow for equivalences occurring on D_{qc} , Perf , or D_{coh}^b to be compared. Comparable results are not available for algebraic spaces in the required generality. Instead, we carefully track the right adjoint of the $\Phi_{\mathcal{J}_{\Delta_f}}$ on fibers.

2.3. Singularity categories. We next apply and extend the preceding theory to singularity categories. For a Noetherian algebraic space X , its singularity category is the Verdier quotient $D_{\text{sg}}(X) := D_{\text{coh}}^b(X)/\text{Perf}(X)$. This originates in Buchweitz's work in the affine setting [9] and was later developed for schemes by Orlov [39].

Derived categories of singular Noetherian spaces are generally less understood than those of smooth spaces. The singularity category isolates why: $D_{\text{sg}}(X) = 0$ if, and only if, X is regular. Its K -theoretic properties have been studied for schemes [40], as have equivalences between singularity categories [23, 24, 34]. In several important examples, such equivalences are induced by integral transforms on D_{coh}^b .

These observations motivate the following objectives, which investigate how the relative Fourier–Mukai theory developed behaves after passage to singularity categories.

Objective B. 1. *Establish openness of the relatively perfect loci of a kernel.*

Objective B. 2. *Characterize when an induced integral transform on D_{sg} is fully faithful or an equivalence in terms of perfectness of cones of the unit/counit in adjunctions.*

Objective B. 3. *Identify when fully faithfulness and equivalence on D_{sg} can be detected fiberwise, and determine the obstructions when the necessary hypotheses fail.*

For a kernel K on $Y_1 \times_S Y_2$, let $\text{RP}_i(K)$ denote the set of points $p \in |S|$ for which the fiber kernel K_p is relatively perfect over $Y_i \times_S \text{Spec}(k)$ with $t: \text{Spec}(k) \rightarrow S$ a representative of p . In **Objective B. 1**, we will prove that these loci are open. In particular, the work in §2 gives independence of the chosen representative. Preliminary work identifies the support-theoretic mechanism underlying this result in key cases. At every point of $\text{RP}_1(K) \cap \text{RP}_2(K)$, the corresponding fiber integral transform Φ_{K_p} preserves both bounded coherent and perfect complexes. Thus, the intersection of the $\text{RP}_i(K)$ records precisely where fiber transforms induce an exact functor on singularity categories. It contains both $\text{fm}(K)$ and $\text{FM}(K)$, which record fully faithfulness and equivalences.

For **Objective B. 2**, suppose that Φ_K admits an adjoint $\Phi_{K'}$. Fully faithfulness of the induced functor on D_{sg} is controlled by whether the cone of the unit $1 \rightarrow \Phi_{K'} \circ \Phi_K$ is perfect for all bounded pseudocoherent complexes, while equivalence also requires similar conditions for the corresponding counit. Preliminary work identifies the failure of perfection for the cones of the unit and counit as the relevant obstruction. This transfers a difficult computation in D_{sg} to a perfectness problem in D_{coh}^b , where the methods of §2 apply.

Finally, **Objective B. 3** determines the extent to which these properties are fiberwise. Unlike fully faithfulness or equivalences for D_{coh}^b , the corresponding properties on D_{sg} need not be invariant under base change along a choice of representative for a closed point. Preliminary examples using projective curves in positive characteristic exhibit failure of both ascent and descent. However, we can show that if for every closed point, there exist some representative for which the base-changed integral transform is an equivalence or fully faithful, then the global integral transform has the same property. These results will identify precise limits of fiberwise Fourier–Mukai methods on D_{sg} .

2.4. Derived invariances. We will study how a relative Fourier–Mukai equivalence constrains the geometry of the fibers. Let $f_i: Y_i \rightarrow S$ be proper flat morphisms of Noetherian algebraic spaces. Suppose that a pseudocoherent kernel K , relatively perfect over both factors, induces an S -linear equivalence $\Phi_K: D_{\text{coh}}^b(Y_1) \rightarrow D_{\text{coh}}^b(Y_2)$.

Our first goal is to prove the following:

Objective C. 1. *f_1 is Gorenstein if, and only if, f_2 is Gorenstein.*

Preliminary work provides two proofs for **Objective C. 1**. The first combines the fiberwise criteria in §2 with Serre duality for proper algebraic spaces over a field. The second adapts the arguments of [44] to develop a relative Serre duality for proper morphisms. In both approaches, Gorensteinness is characterized by invertibility of the

relative dualizing complex, and equivalently to the existence of a relative Serre functor on the category of perfect complexes.

The next goal is to prove the following:

Objective C. 2. f_1 is smooth if, and only if, f_2 is smooth.

Preliminary work has a different route than that of [Objective C. 1](#). First, we establish fiberwise criteria for smoothness of finitely presented flat morphisms of Noetherian algebraic spaces. Although smoothness can be tested after all field valued base changes, we refine by showing it is enough to test on a single representative of each point. Independence of the representative follows by passing to a common field extension and using descent. Then we prove that regularity of a Noetherian algebraic space is characterized categorically by $D_{\text{coh}}^b = \text{Perf}$. This is nontrivial since standard scheme-theoretic arguments do not apply. Combining these together, we can test for geometric regularity on fibers, and prove smoothness.

A long-term goal is to prove an analog of §2 for Cohen–Macaulayness. This appears more subtle, and extends beyond the scope of §2.

3. HIGHER SINGULARITIES

My other research program uses mixed Hodge modules to extract information from D_{coh}^b . Saito’s theory of mixed Hodge modules provides a bridge between topology, Hodge theory, and algebraic geometry [42, 43]. It is centered on the bounded derived category of mixed Hodge modules $D^b(\text{MHM}(X))$ on an embeddable complex variety X .

For each $p \in \mathbb{Z}$, there is an exact graded de Rham functor,

$$\text{gr}_p^F \text{DR}_X: D^b(\text{MHM}(X)) \rightarrow D_{\text{coh}}^b(X).$$

These functors produce important bounded coherent complexes on X . In particular, the *trivial Hodge module* gives the p -th Du Bois complex,

$$\underline{\Omega}_X^p \cong \text{gr}_{-p}^F \text{DR}_X(\mathbb{Q}_X^H[\dim X])[p - \dim X].$$

Du Bois complexes now play a central role in singularity theory [36, 22, 37, 16, 47, 28].

We say that X has *pre- m -Du Bois singularities* if $\mathcal{H}^j(\underline{\Omega}_X^p) = 0$ for all $0 \leq p \leq m$ and $j > 0$. Examples include toric varieties and various varieties with local complete intersection singularities. Parallel constructions using intersection Du Bois complexes describe refinements of rational singularities.

3.1. State of the art. Recent work has connected higher singularities with algebraic K -theory: Shen related higher Du Bois singularities to K -regularity, a measure of $\mathbb{A}^1$ -invariance of algebraic K -groups [46], while Burke-Popa-Shen study negative K -groups through mixed Hodge theory and higher singularities [10].

The p -th graded de Rham functors induce homomorphisms on Grothendieck groups,

$$K_0(\text{gr}_p^F \text{DR}_X): K_0(\text{MHM}(X)) \rightarrow K_0(\text{coh}(X)).$$

No general result shows that the essential image of a fixed graded de Rham functor generates $D_{\text{coh}}^b(X)$, or that the induced morphism on K_0 is surjective. More broadly, no general categorical description of this image is available. It remains unclear what information is encoded by this morphism on K_0 .

This raises the central question of the research program:

Question 1. *To what extent do the graded de Rham images of mixed Hodge modules generate $D_{\text{coh}}^b(X)$, and can the number of extensions required measure the singularities of X ?*

3.2. Beyond the state of the art. In joint work with Venkatesh, I will address [Question 1](#) using geometric methods and generation techniques à la Bondal–Van den Bergh [8]. Standard generation permits shifts, cones, finite coproducts, and direct summands. We seek a stronger dévissage using only shifts and cones (i.e. no summands). This distinction is essential for K -theory. Indeed, because the defining relations in K_0 arise from distinguished triangles, such a dévissage would imply that the class of every coherent sheaf in $K_0(\text{coh}(X))$ lies in the image of a class arising from a mixed Hodge module.

We have three objectives:

Objective D. (Hodge-theoretic dévissage). *Establish that every object of $D_{\text{coh}}^b(X)$ is generated, using only cones and shifts, by the complexes $\mathbf{R}i_* \underline{\Omega}_Z^0$ where $i: Z \rightarrow X$ ranges over integral closed subvarieties. Deduce that the morphism*

$$K_0(\text{gr}_p^F \text{DR}_X): K_0(\text{MHM}(X)) \rightarrow K_0(\text{coh}(X))$$

is surjective for all $p \in \mathbb{Z}$.

Objective E. (Categorical invariants). *Introduce categorical numerics arising from this dévissage and characterize the higher Du Bois singularities detected by their extremal values.*

Objective F. (Bounds). *Establish bounds for fiber products of varieties.*

This research program builds on my prior research with Hodge-theoretic techniques for higher singularities [26], categorical characterizations of singularities [30, 33, 12, 5], and invariants of derived categories for singular varieties [13, 31, 29].

Throughout, a *variety* means a separated reduced (not necessarily irreducible) scheme of finite type over $\mathbb{C}$. Moreover, a variety is *embeddable* if it admits a closed immersion into a smooth variety.

3.3. Hodge-theoretic dévissage. The graded de Rham functors commute with derived pushforward along proper morphisms. More precisely, for every proper morphism $f: Y \rightarrow X$ of varieties, there exists a commutative diagram:

$$\begin{array}{ccc} D^b(\mathrm{MHM}(Y)) & \xrightarrow{\mathrm{gr}_p^F \mathrm{DR}_Y} & D_{\mathrm{coh}}^b(Y) \\ f_*^{\mathrm{MHM}} \downarrow & & \downarrow \mathbf{R}f_* \\ D^b(\mathrm{MHM}(X)) & \xrightarrow{\mathrm{gr}_p^F \mathrm{DR}_X} & D_{\mathrm{coh}}^b(X). \end{array}$$

§3.2 seeks to establish the following strong form of dévissage: *every $E \in D_{\mathrm{coh}}^b(X)$ can be obtained, using only cones and shifts, from the 0-th Du Bois complexes $\mathbf{R}i_*\underline{\Omega}_Z^0$ obtained by pushing forward the zeroth Du Bois complexes of integral closed subvarieties $i: Z \rightarrow X$.* Crucially, direct summands are not used. This places the construction in the setting detected by Grothendieck groups. Our preliminary work provides a tentative proof of this strong dévissage, which is discussed below.

Thomason's criterion [48] translates this dévissage into surjectivity on Grothendieck groups:

Objective D. 1 (Lank–Venkatesh). *Let X be an embeddable variety. For any $p \in \mathbb{Z}$, the induced morphism*

$$K_0(\mathrm{gr}_p^F \mathrm{DR}_X): K_0(\mathrm{MHM}(X)) \rightarrow K_0(\mathrm{coh}(X))$$

is surjective.

This expresses $K_0(\mathrm{coh}(X))$ entirely in terms of classes coming from mixed Hodge modules. Notably, it applies to varieties with arbitrary singularities. Our preliminary proof reduces to $p = 0$ using Tate twists and then proceeds by induction on the Krull dimension. The induction has two key ingredients. First, on the smooth locus $U \subseteq X$, one has $\underline{\Omega}_U^0 \cong \mathcal{O}_U$. Hence, the cone of the natural morphism $\mathcal{O}_X \rightarrow \underline{\Omega}_X^0$ is supported on a proper closed subset. Second, the graded de Rham functors commute with derived pushforwards of proper morphisms. Together, these properties allow for induction on the dimension of the support.

A further input is a coherent dévissage result: the complexes $\mathbf{R}i_*\mathcal{O}_Z$ where $i: Z \rightarrow X$ ranges over all integral subvarieties, generate $D_{\mathrm{coh}}^b(X)$ using only shifts and cones. The available quasi-projective argument uses an ample line bundle but this does not apply to every embeddable variety. Preliminary work develops the required extension in the broader setting of Noetherian algebraic spaces. Although §3.2 concerns complex varieties, this stronger result is of independent interest.

3.4. Categorical numerics. Motivated by this rigid form of generation, we introduce numerics that measure how efficiently objects can be built from mixed Hodge-theoretic data. For an embeddable variety X and $p \in \mathbb{Z}$, set Ψ_X^p to be the collection in $D_{\mathrm{coh}}^b(X)$ containing the zero object and those of the form $\mathbf{R}i_*\underline{\Omega}_Z^p$ where $i: Z \rightarrow X$ is a closed immersion from integral subvarieties.

For $\mathcal{T}$ a triangulated category and $\mathcal{U}, \mathcal{S} \subseteq \mathcal{T}$, define $\mathrm{lex}^{\mathcal{S}}(\mathcal{U})$ as the smallest $n \geq 0$ such that all objects of $\mathcal{U}$ can be built as an n -fold extension of shifted objects of $\mathcal{S}$. If no such n exists, we set $\mathrm{lex}^{\mathcal{S}}(\mathcal{U}) = \infty$. By convention, $\mathrm{lex}^{\mathcal{S}}(\mathcal{U}) = 0$ if $\mathcal{U}$ consists only of the zero object, while $\mathrm{lex}^{\mathcal{S}}(\mathcal{U}) = 1$ if $\mathcal{U}$ consists of shifts of objects of $\mathcal{S}$.

This is related to ‘level’ introduced in [4] but we do not allow direct summands or finite coproducts. Preliminary work has studied basic properties of lex . Our plan is to study values such as $\mathrm{lex}^{\Psi_X^p}(\mathcal{H}^0(\underline{\Omega}_X^p))$ and $\mathrm{lex}^{\Psi_X^0 \oplus \Psi_X^p}(\mathcal{H}^0(\underline{\Omega}_X^p))$. In fact, the preliminary work on **Objective D. 1** implies $\mathrm{lex}^{\Psi_X^0 \oplus \Psi_X^p}(\mathcal{H}^0(\underline{\Omega}_X^p))$ is always finite. Here, $\Psi_X^0 \oplus \Psi_X^p$ means $\{A \oplus B: A \in \Psi_X^0, B \in \Psi_X^p\}$.

The goal of §3.2 is to show that extremal cases recover various classes of singularities.

Objective E. 1 (Lank–Venkatesh). *Let X be an embeddable integral variety and $m \geq 0$.*

- (1) *X has pre- m -Du Bois singularities if, and only if, for all $0 \leq p \leq m$, one has $\text{lct}^{\Psi^p_X}(\mathcal{H}^0(\underline{\Omega}^p_X)) = 1$.*
- (2) *X has Du Bois singularities if, and only if, $\text{lct}^{\Psi^0_X}(\mathcal{O}_X) = 1$.*

Thus, these numerics measure the failure of a variety to have higher Du Bois singularities. Other numerics appearing in [30, 5, 33, 11] use direct summands. In contrast, the numerics here are defined *without* direct summands or finite coproducts.

Preliminary work provides distinct proof strategies for the two parts of **Objective E. 1**. The pre- m -Du Bois argument does not use a splitting criterion, unlike most existing criteria for these singularities. These include [27, Theorem 1], [45, Theorem 4.3], [26, Theorem 3.3], [28, Theorem 10.4], and [49, Theorem F].

By contrast, the Du Bois characterization in **Objective E. 1** currently appears to require a splitting result [45]. Since the two arguments are independent, this dependence does not affect the first part of **Objective E. 1**. An additional objective is to determine whether the splitting input in the second part of **Objective E. 1** can be removed.

3.5. Bounds. Next we look towards producing bounds for fiber products. In particular, a crucial step is to obtain sufficient control over the p -th Du Bois complexes of fiber products. Preliminary work with Dirks has produced the following Künneth decomposition, which serves as initial tool here. Let X and Y be embeddable varieties. Denote by f_X, f_Y the projections from $X \times_{\mathbb{C}} Y$ to X and Y . For any $0 \leq p \leq \dim X + \dim Y$, there is a decomposition $\underline{\Omega}^p_{X \times_{\mathbb{C}} Y} \cong \bigoplus_{s+t=p} \mathbf{L}f_X^* \underline{\Omega}^s_X \otimes^{\mathbf{L}} \mathbf{L}f_Y^* \underline{\Omega}^t_Y$.

The decomposition provides the main input for product bounds. It also implies that the fiber product of two pre- m -Du Bois varieties has pre- m -Du Bois singularities. In particular, this generalizes [14, Theorem 3.9] to the setting of higher singularities. Therefore, it naturally leads to the problem of measuring the failure of $X \times_{\mathbb{C}} Y$ to have such singularities in terms of numerics associated with X and Y .

Preliminary arguments suggest we have a proof of the following:

Objective F. 1. *Let X and Y be embeddable integral varieties. Denote by f_X, f_Y the projections from $X \times_{\mathbb{C}} Y$ to X and Y . Then there are inequalities:*

- (1) $\text{lct}^{\Psi^0_{X \times_{\mathbb{C}} Y}}(\mathcal{O}_{X \times_{\mathbb{C}} Y}) \leq \text{lct}^{\Psi^0_X}(\mathcal{O}_X) \cdot \text{lct}^{\Psi^0_Y}(\mathcal{O}_Y)$
- (2) $\text{lct}^{\Psi^0_{X \times_{\mathbb{C}} Y}}(\mathcal{H}^0(\underline{\Omega}^0_{X \times_{\mathbb{C}} Y})) \leq \text{lct}^{\Psi^0_X}(\mathcal{H}^0(\underline{\Omega}^0_X)) \cdot \text{lct}^{\Psi^0_Y}(\mathcal{H}^0(\underline{\Omega}^0_Y)).$

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